

Vector Calculus

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(adapted from Fieller (2016), [Section 7.2](#))

This section covers derivatives of functions of vectors and matrices. Its linear algebra prerequisites, such as vectors, matrices, transposes, dot products, and quadratic forms, are covered in [Linear Algebra](#).

Let $\tilde{x}$ and $\tilde{\beta}$ be column vectors of length p (see [column vector](#) and [dot product](#)).

Definition 0.1 (Vector derivative). If $f(\tilde{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\tilde{\beta}$, such as $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$, then its **vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) \\ \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) \\ \vdots \\ \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

Video lecture

Hutchinson's [Gradients Refresher](#) (17 min) covers gradients and partial derivatives, the ideas behind this section (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on partial derivatives, the multivariate chain rule, and gradients:

- [What Partial Derivatives Are \(Hands-on Introduction\)](#)
- [Partial Derivative Exercises](#)
- [Advanced Partial Derivatives](#)

- [Advanced Partial-Derivative Exercises](#)
- [Partial Derivative Notation](#)
- [The Chain Rule for Partial Derivatives](#)
- [Exercises on the Multivariate Chain Rule](#)
- [The Gradient of Quadratic Cost](#)
- [The Gradient of Mean Squared Error](#)
- [Higher-Order Partial Derivatives](#)
- [Exercise on Higher-Order Partial Derivatives](#)

Checking a gradient by its shape

The gradient always has the same shape as the input it is taken with respect to, because it holds exactly one partial derivative per component of that input:

Table 1: The shape of a gradient follows the shape of what it differentiates with respect to.

f is a function of	its gradient is
a single number	a single number — the ordinary derivative
a vector in $\mathbb{R}^p$	a vector in $\mathbb{R}^p$
a matrix in $\mathbb{R}^{m \times n}$	a matrix in $\mathbb{R}^{m \times n}$

The table is the cheapest check there is on a gradient just worked out by hand: if the shapes disagree, something went wrong. It is the shape rule of matrix multiplication again, applied to calculus.

The course website’s version of this page has an interactive plot [here](#).

Exercise 0.1 (Take a gradient by hand). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be

$$f(\tilde{w}) = 3w_1w_2 + w_2^3 - 5w_1$$

1. Compute $\partial f / \partial w_1$, treating w_2 as a constant.
2. Compute $\partial f / \partial w_2$, treating w_1 as a constant.
3. Assemble $\nabla_{\tilde{w}} f$ and evaluate it at $\tilde{w} = \begin{bmatrix} 2 & -1 \end{bmatrix}^\top$.
4. Check the shape of your answer against Table [1](#).

Solution

Solution 0.1. **1.** Differentiate with respect to w_1 , holding w_2 fixed. The first term is the constant $3w_2$ times w_1 , so it contributes $3w_2$. The second term has no w_1 in it, so it

contributes 0. The third term is -5 times w_1 , so it contributes -5 :

$$\frac{\partial f}{\partial w_1} = 3w_2 + 0 - 5 = 3w_2 - 5$$

2. Now with respect to w_2 , holding w_1 fixed. The first term is the constant $3w_1$ times w_2 , contributing $3w_1$. The second term contributes $3w_2^2$. The third has no w_2 in it, contributing 0:

$$\frac{\partial f}{\partial w_2} = 3w_1 + 3w_2^2 + 0 = 3w_1 + 3w_2^2$$

3. Stack the two, in the order the coordinates are numbered:

$$\nabla_{\tilde{w}} f(\tilde{w}) = \begin{bmatrix} 3w_2 - 5 \\ 3w_1 + 3w_2^2 \end{bmatrix}$$

The gradient is a vector-valued *function* of $\tilde{w}$, not a single vector. At $\tilde{w} = \begin{bmatrix} 2 & -1 \end{bmatrix}^\top$,

$$\nabla_{\tilde{w}} f = \begin{bmatrix} 3(-1) - 5 \\ 3(2) + 3(-1)^2 \end{bmatrix} = \begin{bmatrix} -3 - 5 \\ 6 + 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$$

4. The input was a vector in $\mathbb{R}^2$ and so is the answer, as Table 1 requires. Note the signs: from this point, increasing w_1 *decreases* f while increasing w_2 increases it, so the uphill direction is neither axis.

Definition 0.2 (Row-vector derivative). If $f(\tilde{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\tilde{\beta}$, such as $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$, then its **row-vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) & \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) & \cdots & \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

Theorem 0.1 (Row and column derivatives are transposes).

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \left(\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) \right)^\top$$

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \left(\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) \right)^\top$$

i Proof

Proof. By Definition 0.1 and Definition 0.2, entry j of both $\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta})$ and $\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta})$ is $\frac{\partial}{\partial \beta_j} f(\tilde{\beta})$; the first is a $p \times 1$ column and the second a $1 \times p$ row with the same entries in the same order, so each is the transpose of the other. $\square$

Example 0.1 (Row and column derivatives of a linear function). For $f(\tilde{\beta}) = 3\beta_1 + 5\beta_2$:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad \frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = [3 \quad 5],$$

and each is the transpose of the other.

Definition 0.3 (Derivative of a vector-valued function). If $\tilde{y} = \tilde{y}(\tilde{\beta}) = (y_1, \dots, y_q)^\top$ is a $q \times 1$ vector-valued function of the $p \times 1$ vector $\tilde{\beta}$, its **derivative with respect to $\tilde{\beta}$** is the $p \times q$ matrix whose (i, j) entry is

$$\left[\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top \right]_{ij} \stackrel{\text{def}}{=} \frac{\partial}{\partial \beta_i} y_j, \quad i = 1, \dots, p, \quad j = 1, \dots, q.$$

These notes use this *denominator layout* throughout: rows index the entries of $\tilde{\beta}$ (the denominator) and columns index the entries of $\tilde{y}$ (the numerator), so column j is the vector derivative $\frac{\partial}{\partial \tilde{\beta}} y_j$ (Definition 0.1). Both $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}$ and $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top$ denote this $p \times q$ matrix.

Example 0.2 (Differentiating a 3×1 function of a 2×1 vector). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ ($p = 2$) and $\tilde{y}(\tilde{\beta}) = (\beta_1^2, \beta_1\beta_2, 3\beta_2)^\top$ ($q = 3$). Then

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top}_{2 \times 3} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} \beta_1^2 & \frac{\partial}{\partial \beta_1} \beta_1\beta_2 & \frac{\partial}{\partial \beta_1} 3\beta_2 \\ \frac{\partial}{\partial \beta_2} \beta_1^2 & \frac{\partial}{\partial \beta_2} \beta_1\beta_2 & \frac{\partial}{\partial \beta_2} 3\beta_2 \end{bmatrix} = \begin{bmatrix} 2\beta_1 & \beta_2 & 0 \\ 0 & \beta_1 & 3 \end{bmatrix}$$

Remark 0.1 (Numerator layout). Some sources use the transposed, *numerator layout*, in which the derivative is the $q \times p$ Jacobian matrix with (j, i) entry $\frac{\partial}{\partial \beta_i} y_j$. For example, in numerator layout, the derivative in Example 0.2 is the 3×2 matrix

$$\begin{bmatrix} 2\beta_1 & 0 \\ \beta_2 & \beta_1 \\ 0 & 3 \end{bmatrix}$$

the transpose of the 2×3 matrix found there. Check a source's layout before combining its formulas with these.

Definition 0.4 (Constant). A $q \times 1$ vector $\tilde{x}$ is **constant with respect to** the $p \times 1$ vector $\tilde{\beta}$ if its derivative (Definition 0.3) is zero:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{p \times q} = \underbrace{\mathbf{0}}_{p \times q}$$

Example 0.3 (A constant vector). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ and $\tilde{x} = (3, 5)^\top$, so $x_1 = 3$ and $x_2 = 5$ do not depend on $\tilde{\beta}$. Expanding $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top$ into its matrix of scalar partial derivatives (Definition 0.3) and evaluating each entry:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{2 \times 2} = \frac{\partial}{\partial \tilde{\beta}} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} x_1 & \frac{\partial}{\partial \beta_1} x_2 \\ \frac{\partial}{\partial \beta_2} x_1 & \frac{\partial}{\partial \beta_2} x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} 3 & \frac{\partial}{\partial \beta_1} 5 \\ \frac{\partial}{\partial \beta_2} 3 & \frac{\partial}{\partial \beta_2} 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \underbrace{\mathbf{0}}_{2 \times 2}$$

Every entry is the derivative of a constant, so $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top = \underbrace{\mathbf{0}}_{2 \times 2}$ and $\tilde{x}$ is constant with respect to $\tilde{\beta}$ (Definition 0.4).

Theorem 0.2 (Derivative of a dot product). If $\tilde{x}$ is constant with respect to $\tilde{\beta}$, then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{x} \cdot \tilde{\beta})}_{p \times 1} = \underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta} \cdot \tilde{x})}_{p \times 1} = \underbrace{\tilde{x}}_{p \times 1}$$

i Proof

Proof.

$$\begin{aligned}\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) &= \begin{bmatrix} \frac{\partial}{\partial \beta_1}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \\ \frac{\partial}{\partial \beta_2}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \\ \vdots \\ \frac{\partial}{\partial \beta_p}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \\ &= \tilde{x}\end{aligned}$$

□

Example 0.4 (Derivative of a dot product). Let $\tilde{x} = (3, 5)^\top$ (constant with respect to $\tilde{\beta}$; see Example 0.3) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{x} \cdot \tilde{\beta} = 3\beta_1 + 5\beta_2$, and by Theorem 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta})}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Verifying entry-wise:

$$\frac{\partial}{\partial \tilde{\beta}}(3\beta_1 + 5\beta_2) = \begin{pmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1 + 5\beta_2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1 + 5\beta_2) \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Both methods agree.

Theorem 0.3 (Product rule for dot-products). *If $\tilde{a} = \tilde{a}(\tilde{x})$ and $\tilde{b} = \tilde{b}(\tilde{x})$ are differentiable $p \times 1$ vector functions of $\tilde{x}$, then:*

$$\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{a}}_{p \times 1} \cdot \underbrace{\tilde{b}}_{p \times 1} = \left(\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{a}^\top}_{1 \times p} \right) \underbrace{\tilde{b}}_{p \times 1} + \left(\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{b}^\top}_{1 \times p} \right) \underbrace{\tilde{a}}_{p \times 1}$$

i Proof

Proof. Entry-wise, for $i = 1, \dots, p$:

$$\begin{aligned}
\left[\frac{\partial}{\partial \tilde{x}} (\tilde{a} \cdot \tilde{b}) \right]_i &= \frac{\partial}{\partial x_i} \sum_{k=1}^p a_k b_k \\
&= \sum_{k=1}^p \left(b_k \frac{\partial}{\partial x_i} a_k + a_k \frac{\partial}{\partial x_i} b_k \right) \\
&= \left[\left(\frac{\partial}{\partial \tilde{x}} \tilde{a}^\top \right) \tilde{b} \right]_i + \left[\left(\frac{\partial}{\partial \tilde{x}} \tilde{b}^\top \right) \tilde{a} \right]_i
\end{aligned}$$

□

Example 0.5 (Example of the dot-product rule). Apply Theorem 0.3 with the vector $\tilde{\beta} = (\beta_1, \beta_2)^\top$ in the role of $\tilde{x}$. Let $\tilde{a}(\tilde{\beta}) = (\beta_1, \beta_1 \beta_2)^\top$ and $\tilde{b}(\tilde{\beta}) = (\beta_2, \beta_1)^\top$. Then:

$$\tilde{a} \cdot \tilde{b} = \beta_1 \cdot \beta_2 + \beta_1 \beta_2 \cdot \beta_1 = \beta_1 \beta_2 + \beta_1^2 \beta_2$$

By direct calculation:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} (\tilde{a} \cdot \tilde{b}) = \frac{\partial}{\partial \tilde{\beta}} (\beta_1 \beta_2 + \beta_1^2 \beta_2) = \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1 + \beta_1^2 \end{pmatrix}$$

By the product rule (Theorem 0.3), using $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{a}^\top}_{2 \times 2} = \begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}$ and $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{b}^\top}_{2 \times 2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} (\tilde{a} \cdot \tilde{b}) &= \underbrace{\begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_2 \\ \beta_1 \end{pmatrix}}_{2 \times 1} + \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_1 \\ \beta_1 \beta_2 \end{pmatrix}}_{2 \times 1} \\
&= \begin{pmatrix} \beta_2 + \beta_1 \beta_2 \\ \beta_1^2 \end{pmatrix} + \begin{pmatrix} \beta_1 \beta_2 \\ \beta_1 \end{pmatrix} \\
&= \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1^2 + \beta_1 \end{pmatrix}
\end{aligned}$$

Both methods agree.

Theorem 0.4 (Derivative of a linear map). If $\mathbf{A}$ is an $m \times p$ matrix that is constant with respect to $\tilde{\beta}$, then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{p \times m} (\mathbf{A} \tilde{\beta}) = \underbrace{\mathbf{A}^\top}_{p \times m}$$

i Proof

Proof. For entry (i, j) , where row i indexes the denominator $\tilde{\beta}$ (see Definition 0.3) and column j indexes the numerator $\mathbf{A}\tilde{\beta}$:

$$\begin{aligned} \left[\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{\beta}) \right]_{ij} &= \frac{\partial}{\partial \beta_i} (\mathbf{A}\tilde{\beta})_j \\ &= \frac{\partial}{\partial \beta_i} \sum_{k=1}^p a_{jk} \beta_k \\ &= a_{ji} \\ &= [\mathbf{A}^\top]_{ij} \end{aligned}$$

□

Example 0.6 (Derivative of a linear map). Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ (1×2) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{A}\tilde{\beta} = 2\beta_1 + 3\beta_2$, and by Theorem 0.4:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{\beta})}_{2 \times 1} = \underbrace{\mathbf{A}^\top}_{2 \times 1} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Theorem 0.5 (Vector-derivative of a matrix-vector product). *If $\mathbf{A}$ is an $m \times q$ matrix that is constant with respect to $\tilde{\beta}$, and $\tilde{v} = \tilde{v}(\tilde{\beta})$ is a $q \times 1$ vector that depends on the $p \times 1$ vector $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{v})}_{p \times m} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{A}^\top}_{q \times m}$$

i Proof

Proof. For entry (i, j) , where row i indexes the denominator $\tilde{\beta}$ and column j indexes the numerator $\mathbf{A}\tilde{v}$ (see Definition 0.3):

$$\begin{aligned}
\left[\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{v}) \right]_{ij} &= \frac{\partial}{\partial \beta_i} (\mathbf{A} \tilde{v})_j \\
&= \frac{\partial}{\partial \beta_i} \sum_{k=1}^q a_{jk} v_k \\
&= \sum_{k=1}^q a_{jk} \frac{\partial}{\partial \beta_i} v_k \\
&= \sum_{k=1}^q \left[\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right]_{ik} [\mathbf{A}^\top]_{kj} \\
&= \left[\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{A}^\top \right]_{ij}
\end{aligned}$$

□

Example 0.7 (Vector-derivative of a matrix-vector product). Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ (1×2 , constant) and $\tilde{v}(\tilde{\beta}) = (\beta_1^2, \beta_2^2)^\top$. Then $\mathbf{A} \tilde{v} = 2\beta_1^2 + 3\beta_2^2$. By Theorem 0.5:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{v})}_{2 \times 1} &= \begin{pmatrix} 2\beta_1 & 0 \\ 0 & 2\beta_2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \\
&= \begin{pmatrix} 4\beta_1 \\ 6\beta_2 \end{pmatrix}
\end{aligned}$$

Remark 0.2 (The derivative of a linear map as a special case). This result generalizes Theorem 0.4, which is the special case $\tilde{v} = \tilde{\beta}$ (so that $q = p$, $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}_p$, and $\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{\beta}) = \mathbf{I}_p \mathbf{A}^\top = \mathbf{A}^\top$). For example, Example 0.6 is the case $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ and $\tilde{v} = \tilde{\beta} = (\beta_1, \beta_2)^\top$, where this result gives $\mathbf{I}_2 \mathbf{A}^\top = (2, 3)^\top$.

Theorem 0.6 (Vector-derivative of a product of matrices). If $\mathbf{A}$ ($\ell \times m$) and $\mathbf{B}$ ($m \times q$) are constant with respect to $\tilde{\beta}$, and $\tilde{v} = \tilde{v}(\tilde{\beta})$ is a $q \times 1$ vector that depends on the $p \times 1$ vector $\tilde{\beta}$, then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \mathbf{B} \tilde{v})}_{p \times \ell} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{B}^\top}_{q \times m} \underbrace{\mathbf{A}^\top}_{m \times \ell}$$

i Proof

Proof. Apply Theorem 0.5 with the constant $\ell \times q$ matrix $\mathbf{AB}$, then use the [transpose of a product](#):

$$\begin{aligned}\frac{\partial}{\partial \tilde{\beta}}(\mathbf{AB}\tilde{v}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) (\mathbf{AB})^\top \quad (\text{derivative of a matrix-vector product}) \\ &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{B}^\top \mathbf{A}^\top \quad (\text{transpose of a product})\end{aligned}$$

□

Example 0.8. Let $\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ (1×2), $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2), and $\tilde{v}(\tilde{\beta}) = \tilde{\beta}$ where $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{AB}\tilde{v} = 2\beta_1$, and:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{AB}\tilde{v})}_{2 \times 1} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right)}_{2 \times 2} \underbrace{\mathbf{B}^\top \mathbf{A}^\top}_{2 \times 2 \times 1} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Corollary 0.1 (Derivative of a dot product, transpose-product form). *If $\tilde{x}$ is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta})}_{p \times 1} = \tilde{x}_{p \times 1}$$

i Proof

Proof. Using Theorem 0.2:

Since $\tilde{x}^\top \tilde{\beta} = \tilde{x} \cdot \tilde{\beta}$ (see [dot product](#)), and $\tilde{x}$ is constant with respect to $\tilde{\beta}$:

$$\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta}) = \frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) = \tilde{x}$$

by Theorem 0.2.

□

i Proof

Proof. Using **Theorem 0.5**:

Since $\tilde{x}$ is constant with respect to $\tilde{\beta}$, $\mathbf{A} = \tilde{x}^\top$ is a constant $1 \times p$ matrix. Applying **Theorem 0.5** with $\tilde{v} = \tilde{\beta}$ (so $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}} (\tilde{x}^\top \tilde{\beta}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right) (\tilde{x}^\top)^\top \\ &= \mathbf{I} \cdot \tilde{x} \\ &= \tilde{x} \end{aligned}$$

□

Example 0.9 (Derivative of a transpose product). Let $\tilde{x} = (3, 5)^\top$ and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{x}^\top \tilde{\beta} = 3\beta_1 + 5\beta_2$, and by **Corollary 0.1**:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \left(\underbrace{\tilde{x}^\top}_{1 \times 2} \underbrace{\tilde{\beta}}_{2 \times 1} \right)}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Remark 0.3 (The coefficient gets transposed). This vector derivative formula looks a lot like non-vector calculus, except that you have to transpose the coefficient: in scalar calculus $\frac{\partial}{\partial x}(cx) = c$, but here the coefficient $\tilde{x}^\top$ (a row vector) becomes $\tilde{x}$ (a column vector) in the result. For example, with $\tilde{x} = (2, -1)^\top$, $\tilde{x}^\top \tilde{\beta} = 2\beta_1 - \beta_2$, whose vector derivative is the column vector $(2, -1)^\top = \tilde{x}$, not the row vector $\tilde{x}^\top = (2, -1)$.

Theorem 0.7 (Derivative of a quadratic form). *For a quadratic form (see [quadratic form](#)), if $\mathbf{S}$ is a symmetric $p \times p$ matrix that is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{p \times 1} = \underbrace{2\mathbf{S} \tilde{\beta}}_{p \times 1}$$

i Proof

Proof. Expanding entry-wise, $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k$. Differentiating component-wise with respect to β_i for $i = 1, \dots, p$:

$$\begin{aligned}
\left[\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) \right]_i &= \frac{\partial}{\partial \beta_i} \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k \quad (\text{expand quadratic form}) \\
&= \sum_{k=1}^p s_{ik} \beta_k + \sum_{j=1}^p s_{ji} \beta_j \quad (\text{product rule for } \beta_i \beta_k) \\
&= [\mathbf{S} \tilde{\beta}]_i + [\mathbf{S}^\top \tilde{\beta}]_i \quad (\text{matrix-vector multiplication definition}) \\
&= [(\mathbf{S} + \mathbf{S}^\top) \tilde{\beta}]_i \quad (\text{linearity of matrix multiplication})
\end{aligned}$$

When $\mathbf{S}$ is symmetric ($\mathbf{S} = \mathbf{S}^\top$), $\mathbf{S} + \mathbf{S}^\top = 2\mathbf{S}$, so $\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) = 2\mathbf{S} \tilde{\beta}$. $\square$

Remark 0.4 (Like the derivative of cx^2). This operation is like taking the derivative of cx^2 with respect to x in non-vector calculus: $\frac{\partial}{\partial x}(cx^2) = 2cx$, and Theorem 0.7 says $\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) = 2\mathbf{S} \tilde{\beta}$. For example, with $p = 1$, $\mathbf{S} = (3)$, and $\tilde{\beta} = (\beta_1)$, $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = 3\beta_1^2$, and its derivative is $6\beta_1 = 2\mathbf{S} \tilde{\beta}$.

Example 0.10 (Derivative of a quadratic form). Let $\mathbf{S} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ (2×2 , symmetric and constant) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = 3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2$. By Theorem 0.7:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{2 \times 1} = 2\mathbf{S} \tilde{\beta} = 2 \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Differentiating component-wise directly:

$$\begin{pmatrix} \frac{\partial}{\partial \beta_1} (3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \\ \frac{\partial}{\partial \beta_2} (3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Both methods agree.

Corollary 0.2 (Derivative of a simple quadratic form).

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \tilde{\beta})}_{p \times 1} = \underbrace{2\tilde{\beta}}_{p \times 1}$$

i Proof

Proof. Applying Theorem 0.7 with $\mathbf{S} = \mathbf{I}_{p \times p}$ (which is symmetric and constant with respect to $\tilde{\beta}$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta}) &= \frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{I}_{p \times p} \tilde{\beta}) \quad (\text{rewrite with identity matrix}) \\ &= 2\mathbf{I}_{p \times p} \tilde{\beta} \quad (\text{derivative of a quadratic form, with } \mathbf{S} = \mathbf{I}_{p \times p}) \\ &= 2\tilde{\beta} \quad (\text{identity matrix property}) \end{aligned}$$

□

Remark 0.5 (Like the derivative of x^2). This vector derivative is like taking the derivative of x^2 : in scalar calculus $\frac{\partial}{\partial x} x^2 = 2x$, and Corollary 0.2 says $\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta}) = 2\tilde{\beta}$. For example, with $p = 1$ and $\tilde{\beta} = (\beta_1)$, $\tilde{\beta}^\top \tilde{\beta} = \beta_1^2$, and its derivative is $2\beta_1 = 2\tilde{\beta}$.

Example 0.11 (Derivative of a sum of squares). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$, so $\tilde{\beta}^\top \tilde{\beta} = \beta_1^2 + \beta_2^2$. By Corollary 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta})}_{2 \times 1} = 2\tilde{\beta} = \begin{pmatrix} 2\beta_1 \\ 2\beta_2 \end{pmatrix}$$

Direct partial differentiation yields the same column vector.

Theorem 0.8 (Vector chain rule). Let $\tilde{x}$ be a $p \times 1$ vector, let $\tilde{y} = \tilde{g}(\tilde{x})$ be a $q \times 1$ vector-valued function of $\tilde{x}$, and let $z = f(\tilde{y})$ be a scalar-valued function of $\tilde{y}$, where $\tilde{g}$ and f have continuous partial derivatives. Then $z = f(\tilde{g}(\tilde{x}))$, as a function of $\tilde{x}$, satisfies

$$\underbrace{\frac{\partial z}{\partial \tilde{x}}}_{p \times 1} = \underbrace{\frac{\partial \tilde{y}}{\partial \tilde{x}}}_{p \times q} \underbrace{\frac{\partial z}{\partial \tilde{y}}}_{q \times 1}$$

where $\frac{\partial \tilde{y}}{\partial \tilde{x}}$ is the derivative of Definition 0.3 and $\frac{\partial z}{\partial \tilde{x}}$ and $\frac{\partial z}{\partial \tilde{y}}$ are vector derivatives (Definition 0.1).

Remark 0.6 (The order of the factors matters). The vector chain rule (Theorem 0.8) is like the univariate chain rule, but the order matters now: $\frac{\partial \tilde{y}}{\partial \tilde{x}}$ is $p \times q$ and $\frac{\partial z}{\partial \tilde{y}}$ is $q \times 1$, so the product $\frac{\partial z}{\partial \tilde{y}} \frac{\partial \tilde{y}}{\partial \tilde{x}}$ in the other order is not even defined unless $p = 1$. The version presented here is for the **gradient** (column vector); the **total derivative** (row

vector) would be the **transpose of the gradient**, $(\frac{\partial z}{\partial \tilde{x}})^\top = (\frac{\partial z}{\partial \tilde{y}})^\top (\frac{\partial \tilde{y}}{\partial \tilde{x}})^\top$, with the factors in the reverse order.

Example 0.12 (Applying the vector chain rule). Let $\tilde{x} = (x_1, x_2)^\top$, $\tilde{y} = \tilde{g}(\tilde{x}) = (x_1 + x_2, x_1 x_2)^\top$, and $z = f(\tilde{y}) = y_1^2 + y_2$. Then

$$\begin{aligned} \frac{\partial z}{\partial \tilde{x}} &= \frac{\partial \tilde{y}}{\partial \tilde{x}} \frac{\partial z}{\partial \tilde{y}} && \text{(vector chain rule)} \\ \underset{2 \times 1}{\frac{\partial z}{\partial \tilde{x}}} &= \underset{2 \times 2}{\frac{\partial \tilde{y}}{\partial \tilde{x}}} \underset{2 \times 1}{\frac{\partial z}{\partial \tilde{y}}} && \text{(differentiate } \tilde{g} \text{ and } f) \\ &= \begin{bmatrix} 1 & x_2 \\ 1 & x_1 \end{bmatrix} \begin{bmatrix} 2y_1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2(x_1 + x_2) + x_2 \\ 2(x_1 + x_2) + x_1 \end{bmatrix} && \text{(multiply, and substitute } y_1 = x_1 + x_2) \\ &= \begin{bmatrix} 2x_1 + 3x_2 \\ 3x_1 + 2x_2 \end{bmatrix} && \text{(collect terms)} \end{aligned}$$

This matches differentiating z directly: $z = (x_1 + x_2)^2 + x_1 x_2 = x_1^2 + 3x_1 x_2 + x_2^2$, so $\frac{\partial}{\partial x_1} z = 2x_1 + 3x_2$ and $\frac{\partial}{\partial x_2} z = 3x_1 + 2x_2$. The product in the other order, a 2×1 matrix times a 2×2 matrix, is not defined. The total derivative is the transpose, the row vector $(2x_1 + 3x_2, 3x_1 + 2x_2)$.

See <https://quickfem.com/finite-element-analysis/>, specifically https://quickfem.com/wp-content/uploads/IFEM.AppF_.pdf

See also https://en.wikipedia.org/wiki/Gradient#Relationship_with_Fr%C3%A9chet_derivative

Corollary 0.3 (Vector chain rule for quadratic forms). *If $\tilde{\varepsilon} = \tilde{\varepsilon}(\tilde{\beta})$ is an $n \times 1$ vector-valued function of the $p \times 1$ vector $\tilde{\beta}$, with continuous partial derivatives, then*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\varepsilon}(\tilde{\beta}) \cdot \tilde{\varepsilon}(\tilde{\beta}))}_{p \times 1} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon}(\tilde{\beta}) \right)}_{p \times n} \underbrace{(2\tilde{\varepsilon}(\tilde{\beta}))}_{n \times 1}$$

i Proof

Proof. Apply Theorem 0.8 with $\tilde{x} = \tilde{\beta}$, $\tilde{y} = \tilde{\varepsilon}$, $q = n$, and $z = f(\tilde{\varepsilon}) = \tilde{\varepsilon} \cdot \tilde{\varepsilon} = \tilde{\varepsilon}^\top \tilde{\varepsilon}$:

$$\begin{aligned}
\frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) \frac{\partial}{\partial \tilde{\varepsilon}}(\tilde{\varepsilon}^\top \tilde{\varepsilon}) \quad (\text{vector chain rule}) \\
&= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) (2\tilde{\varepsilon}) \quad (\text{derivative of a simple quadratic form, in } \tilde{\varepsilon})
\end{aligned}$$

The second step is Corollary 0.2, applied to the $n \times 1$ vector $\tilde{\varepsilon}$ in place of $\tilde{\beta}$. $\square$

Example 0.13 (Derivative of the residual sum of squares). Let $\tilde{y}$ ($n \times 1$) and $\mathbf{X}$ ($n \times p$) be constant with respect to $\tilde{\beta}$, and let $\tilde{\varepsilon}(\tilde{\beta}) = \tilde{y} - \mathbf{X}\tilde{\beta}$ be the vector of residuals. By Theorem 0.4, $\frac{\partial}{\partial \tilde{\beta}}(\mathbf{X}\tilde{\beta}) = \mathbf{X}^\top$, and $\frac{\partial}{\partial \tilde{\beta}}\tilde{y} = \mathbf{0}_{p \times n}$ because $\tilde{y}$ is constant, so $\frac{\partial}{\partial \tilde{\beta}}\tilde{\varepsilon} = -\mathbf{X}^\top$ ($p \times n$). By Corollary 0.3:

$$\begin{aligned}
\frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= (-\mathbf{X}^\top)(2\tilde{\varepsilon}) \quad (\text{vector chain rule for quadratic forms}) \\
&= -2\mathbf{X}^\top(\tilde{y} - \mathbf{X}\tilde{\beta}) \quad (\text{substitute } \tilde{\varepsilon} = \tilde{y} - \mathbf{X}\tilde{\beta})
\end{aligned}$$

Setting this $p \times 1$ vector to $\tilde{0}$ gives the normal equations $\mathbf{X}^\top \mathbf{X} \tilde{\beta} = \mathbf{X}^\top \tilde{y}$ of least squares.

Definition 0.5 (Matrix derivative). For a scalar-valued function $f(\mathbf{X})$ of an $m \times n$ matrix $\mathbf{X}$, the **matrix derivative** is the $m \times n$ matrix whose (i, j) entry is the partial derivative of f with respect to the (i, j) entry of $\mathbf{X}$:

$$\left[\frac{\partial}{\partial \mathbf{X}} f \right]_{ij} = \frac{\partial}{\partial X_{ij}} f$$

Example 0.14 (The matrix derivative of a trace). Let $\mathbf{X}$ be a 2×2 matrix and $f(\mathbf{X}) = \text{tr}(\mathbf{X}) = X_{11} + X_{22}$ (see [trace](#)). Then $\frac{\partial}{\partial X_{ij}} f = 1$ if $i = j$ and 0 otherwise, so:

$$\frac{\partial}{\partial \mathbf{X}} f = \mathbf{I}_2$$

Theorem 0.9 (Matrix derivative of the trace of a matrix product). If $\mathbf{A}$ ($r \times m$) and $\mathbf{B}$ ($n \times r$) are constant with respect to the $m \times n$ matrix $\mathbf{X}$, then:

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{A}\mathbf{X}\mathbf{B})}_{m \times n} = \underbrace{\mathbf{A}^\top}_{m \times r} \underbrace{\mathbf{B}^\top}_{r \times n}$$

i Proof

Proof. Write A_{kl} , X_{kl} , and B_{kl} for the entries of $\mathbf{A}$, $\mathbf{X}$, and $\mathbf{B}$. For entry (i, j) :

$$\begin{aligned}
 \left[\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{A}\mathbf{X}\mathbf{B}) \right]_{ij} &= \frac{\partial}{\partial X_{ij}} \sum_{a=1}^r \sum_{b=1}^m \sum_{c=1}^n A_{ab} X_{bc} B_{ca} \quad (\text{trace of the } r \times r \text{ product } \mathbf{A}\mathbf{X}\mathbf{B}) \\
 &= \sum_{a=1}^r A_{ai} B_{ja} \quad (\text{only the terms with } b = i, c = j \text{ depend on } X_{ij}) \\
 &= \sum_{a=1}^r [\mathbf{A}^\top]_{ia} [\mathbf{B}^\top]_{aj} \quad (\text{definition of the transpose}) \\
 &= [\mathbf{A}^\top \mathbf{B}^\top]_{ij} \quad (\text{definition of matrix multiplication})
 \end{aligned}$$

□

Remark 0.7 (Why the theorem uses the trace). The trace makes $\text{tr}(\mathbf{A}\mathbf{X}\mathbf{B})$ a scalar, so its matrix derivative (Definition 0.5) is again an $m \times n$ matrix. The matrix product $\mathbf{A}\mathbf{X}\mathbf{B}$ itself (without the trace) is an $r \times r$ matrix, and each of its r^2 entries has a partial derivative with respect to each of the mn entries of $\mathbf{X}$. Those $r^2 mn$ partial derivatives form a four-index array (a fourth-order tensor), not a matrix, which is why this result is stated for the scalar $\text{tr}(\mathbf{A}\mathbf{X}\mathbf{B})$.

For example, with $\mathbf{A} = \mathbf{B} = \mathbf{I}_2$, the product $\mathbf{A}\mathbf{X}\mathbf{B} = \mathbf{X}$ has 4 entries, each with 4 partial derivatives, 16 in all, while its trace $X_{11} + X_{22}$ has the 4 partial derivatives that form the 2×2 matrix $\mathbf{I}_2$ of Example 0.14.

Example 0.15 (Differentiating a weighted trace). Let $\mathbf{A} = \mathbf{I}_2$ (2×2) and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2). Then $\text{tr}(\mathbf{A}\mathbf{X}\mathbf{B}) = 2X_{11} + 3X_{22}$, and:

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{A}\mathbf{X}\mathbf{B})}_{2 \times 2} = \underbrace{\mathbf{A}^\top}_{2 \times 2} \underbrace{\mathbf{B}^\top}_{2 \times 2} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

Exercise 0.2 (A gradient that does not mention its variable). Fix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix}$$

let $W \in \mathbb{R}^{2 \times 3}$ vary, and define

$$g(W) = \sum_{i=1}^2 \sum_{j=1}^3 W_{ij} A_{ij}$$

which is the entry-by-entry version of the inner product for matrices. Compute $\nabla_W g(W)$, and say what is unusual about the answer.

Solution

Solution 0.2. Written out over the six positions,

$$g(W) = 2W_{11} - W_{12} + 0W_{13} + 4W_{21} + 3W_{22} - 2W_{23}$$

Each W_{ij} appears in exactly one term, multiplied by A_{ij} and by nothing else, so differentiating with respect to it leaves A_{ij} behind:

$$\frac{\partial g}{\partial W_{ij}} = A_{ij}$$

Collecting those partial derivatives into a matrix of the same shape as W , as Table 1 requires,

$$\nabla_W g(W) = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix} = A$$

What is unusual is that the gradient came out **constant**: it does not mention W at all. The constant answer is not a quirk of this particular A . g is a linear function of W , and the gradient of a linear function is constant everywhere, for the same reason the derivative of $f(w) = cw$ is c no matter where it is evaluated.

A constant gradient is the easy case, and it is not the case we usually face. Most objectives are curved — the squared error of a linear model is the standard example — so their gradient changes from point to point and the downhill direction has to be worked out afresh at every step.

Further reading

- Marsden and Tromba (2013) is a standard textbook on multivariable and vector calculus. It covers differentiation of functions of several variables, multiple integrals, line and surface integrals, and the theorems of Green, Gauss, and Stokes.

See also the [Linear Algebra and Vector Calculus further reading](#).

- [Hua Zhou's lecture notes for "UCLA Biostat 216 - Mathematical Methods for Biostatistics" \(2023 Fall\)](#)

References

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